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Liquid cooling device

Abstract

A liquid cooling device includes a water cooling radiator, a first pump and a cold plate. The water cooling radiator has a first surface and a second surface, the first surface and the second surface are located on opposite sides of the water cooling radiator, the first pump is disposed on the first surface or the second surface of the water cooling radiator, and the cold plate is disposed on the second surface of the water cooling radiator.

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Background/Summary

RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Application Ser. No. 63/280,870, filed Nov. 18, 2021 and U.S. Provisional Application Ser. No. 63/303,263, filed Jan. 26, 2022, the disclosures of which are incorporated herein by reference in their entireties.

TECHNICAL FIELD
(1) The present disclosure generally relates to a liquid cooling device. More particularly, the present disclosure relates to a liquid cooling device of a display card.
BACKGROUND
(2) With the advancement of technology, electronic products have become more popular, and gradually changed the life or work of many people. As the speed of the computers increases, the

calculating power of the display cards becomes more and more powerful so that the temperature control of the electronic components such as the graphics processing units (GPUs) is more important.
(3) Electronic components such as the graphics processing units generate heat during operation and require proper cooling to achieve the best performance. In order to keep the graphics processing unit and other electronic components operating at a proper temperature, a liquid cooling device or an air cooling device is usually used.
(4) In the current water-cooling heat dissipation device, the working fluid flows into the cold plate through the pipeline, and the cold plate contacts the surface of the electronic component, e.g. the graphics processing unit, to take away the heat generated by the electronic component during operation, thereby reducing the operating temperature of the electronic component, and improving the working efficiency of the display cards.
(5) Therefore, there is a need to improve the performance and efficiency of the water-cooling heat dissipation devices so as to improve the performance and efficiency of the display cards, which is also a challenge currently faced by those skilled in the art.

SUMMARY

(6) One objective of the embodiments of the present invention is to provide a liquid cooling device to effectively improve the cooling efficiency thereof as well as effectively improve the performance and efficiency of the electronic devices such as the display cards.
(7) To achieve these and other advantages and in accordance with the objective of the embodiments of the present invention, as the embodiment broadly describes herein, the embodiments of the present invention provides a liquid cooling device, a water cooling radiator, a first pump, and a cold plate. The water cooling radiator includes a first surface and a second surface, and the first surface and the second surface are respectively located on opposite sides of the water cooling radiator. The first pump is disposed on the first surface or the second surface of the water cooling radiator, and the cold plate is disposed on the second surface of the water cooling radiator.
(8) In some embodiments, the water cooling radiator further includes a first water tank, a second water tank, a third water tank, a plurality of heat radiating fins and a plurality of water flow paths. The heat radiating fins are disposed between the first water tank, the second water tank and the third water tank, and the water flow paths pass through in the heat radiating fins.
(9) In some embodiments, the liquid cooling device further includes a second pump, and the first pump is fixed and communicated with the first water tank and the second pump is fixed and communicated with the second water tank.
(10) In some embodiments, the liquid cooling device further includes at least one cooling fan disposed on the first surface or the second surface of the water cooling radiator, and located adjacent to the first pump.
(11) In some embodiments, the liquid cooling device further includes a first cooling fan and a second cooling fan. The first cooling fan is disposed on the first surface or the second surface of the water cooling radiator, and located adjacent to the first pump. In addition, the second cooling fan is disposed on the first surface or the second surface of the water cooling radiator, and located adjacent to the second pump.
(12) In some embodiments, the first pump and the first cooling fan are disposed on the first surface of the water cooling radiator, and the second pump and the second cooling fan are also disposed on the first surface of the water cooling radiator.
(13) In some embodiments, the first pump and the first cooling fan are disposed on the second surface of the water cooling radiator, and the second pump and the second cooling fan are also disposed on the second surface of the water cooling radiator.
(14) In some embodiments, the first pump and the first cooling fan are disposed on the second surface of the water cooling radiator, and the second pump and the second cooling fan are disposed on the first surface of the water cooling radiator. In addition, two ends of the water cooling radiator

include a step.
(15) In some embodiments, the liquid cooling device further includes a third pump fixed and communicated with the third water tank. The first pump, the first cooling fan, the second pump, the second cooling fan and the third pump are disposed on the first surface of the water cooling radiator, and the first cooling fan is located between the first pump and the third pump, and the second cooling fan is located between the second pump and the third pump.
(16) In some embodiments, the first pump is fixed and communicated with the third water tank, and the first pump is disposed on the first surface of the water cooling radiator, and located between a first cooling fan and a second cooling fan.
(17) In some embodiments, the third water tank includes a first partition plate and a second partition plate. The second partition plate and the first partition plate separate the third water tank into three areas, wherein a first water inlet and a first water outlet of the cold plate are respectively connected to two corresponding areas of the three areas, and another area of the three areas is connected to the first pump and disconnected from the cold plate.
(18) In some embodiments, the heat radiating fins include a plurality of first heat radiating fins, a plurality of second heat radiating fins and a plurality of third heat radiating fins. The second heat radiating fins are formed between the first heat radiating fins and the third heat radiating fins, and the first heat radiating fins, the second heat radiating fins and the third heat radiating fins have different intervals.
(19) In some embodiments, an interval of the first heat radiating fins is smaller than an interval of the second heat radiating fins, and the interval of the second heat radiating fins is smaller than an interval of the third heat radiating fins.
(20) In some embodiments, the cold plate is connected to the third water tank.
(21) In some embodiments, the third water tank includes a first partition plate to separate the third water tank into two areas.
(22) In some embodiments, the cold plate includes a first chamber and first skived fins. The first chamber includes a first water inlet and a first water outlet, and the first water inlet and the first water outlet are respectively connected to corresponding areas of the two areas of the third water tank. In addition, the first skived fins are disposed in the first chamber, and located between the first water inlet and the first water outlet.
(23) In some embodiments, the third water tank further includes a second partition plate, and the second partition plate and the first partition plate are formed a cross partition plate to separate the third water tank into four areas. In addition, the cold plate further includes a second chamber, a chamber partition plate and second skived fins. The second chamber includes a second water inlet and a second water outlet, and the first water inlet, the first water outlet, the second water inlet and the second water outlet are respectively connected to corresponding areas of the four areas of the third water tank. The chamber partition plate is formed between the first chamber and the second chamber to isolate the first chamber and the second chamber. In addition, the second skived fins is disposed in the second chamber, and located between the second water inlet and the second water outlet.
(24) Hence, the aforementioned liquid cooling device can provide a larger heat dissipation area of the water cooling radiator without increasing the length, width and height of the liquid cooling device, and further improve the heat dissipation efficiency of the liquid cooling device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The foregoing aspects and many of the attendant advantages of this invention will be more readily appreciated as the same becomes better understood by reference to the following detailed

description, when taken in conjunction with the accompanying drawings, wherein:
(2) FIG. **1** illustrates a schematic view showing a liquid cooling device according to one embodiment of the present invention;
(3) FIG. **2** illustrates a schematic exploded view of the liquid cooling device of FIG. **1**;
(4) FIG. **3** illustrates a schematic exploded view showing a liquid cooling device according to another embodiment of the present invention;
(5) FIG. **4** illustrates a schematic exploded view showing a liquid cooling device according to further another embodiment of the present invention;
(6) FIG. **5** illustrates a schematic exploded view showing a liquid cooling device according to still further another embodiment of the present invention;
(7) FIG. **6** illustrates a schematic view showing a bottom plate of a cold plate of the liquid cooling device of FIG. **5**;
(8) FIG. **7** illustrates a schematic view showing various configuration embodiments of cooling fans, water cooling radiators and pumps of the liquid cooling device; and
(9) FIG. **8** illustrates a schematic view showing a water cooling radiator of a liquid cooling device according to one embodiment of the present invention.
DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT
(10) The following description is of the best presently contemplated mode of carrying out the present disclosure. This description is not to be taken in a limiting sense but is made merely for the purpose of describing the general principles of the invention. The scope of the invention should be determined by referencing the appended claims.
(11) FIG. **1** illustrates a schematic view of a liquid cooling device according to one embodiment of the present invention, and FIG. **2** illustrates a schematic exploded view thereof. FIG. **3** illustrates another embodiment of a liquid cooling device according to the present invention, FIG. **4** illustrates further another embodiment thereof, FIG. **5** illustrates still further another embodiment thereof, FIG. **6** illustrates a bottom plate of a cold plate of the liquid cooling device of FIG. **5**. In addition, FIG. **7** illustrates various configuration embodiments of cooling fans, water cooling radiators and pumps of the liquid cooling device, and FIG. **8** illustrates another embodiment of the water cooling radiator of the liquid cooling device.
(12) Refer to FIG. **1**, the liquid cooling device **100** includes a water cooling radiator **130**, at least one first pump **110** and a cold plate **140**. The water cooling radiator **130** includes a first surface **101** and a second surface **102**, and the first surface **101** and the second surface **102** are respectively located on opposite sides of the water cooling radiator **130**. In addition, the cold plate **140** is disposed on the second surface **102** of the water cooling radiator **130**. That is to say, the cold plate **140** is adjacent to the second surface **102** of the water cooling radiator **130**, and the first surface **101** is another surface of the water cooling radiator **130** opposite to the cold plate **140**.
(13) In some embodiments, the first pump **110** is disposed on the first surface **101** of the water cooling radiator **130**. In some embodiment, the first pump **110** can be disposed on the second surface **102** of the water cooling radiator **130**, without departing from the spirit and scope of the present invention.
(14) In some embodiments, the water cooling radiator **130** further includes a first water tank **132**, a second water tank **134**, a third water tank **136**, a plurality of heat radiating fins **138** and a plurality of water flow paths **139**. The first water tank **132** is disposed on the left side of the water cooling radiator **130** in FIG. **1**, the second water tank **134** is disposed on the right side of the water cooling radiator **130**, and the third water tank **136** is located between the first water tank **132** and the second water tank **134**. The heat radiating fins **138** are disposed between the first water tank **132**, the second water tank **134** and the third water tank **136**, and the water flow paths **139** pass through in the heat radiating fins **138** so as to carry the heat in the working fluid through the water flow paths **139** to the heat radiating fins **138**, and remove the heat by cooling fans.
(15) In some embodiments, the water flow paths **139** are flat water flow pipes to increase the

contact area with the heat radiating fins **138** and reduce the shading area affecting the heat radiating fins **138**.
(16) In some embodiments, the liquid cooling device **100** further includes a second pump **120**, and the first pump **110** is fixed and communicated with the first water tank **132**, and the second pump **120** is fixed and communicated with the second water tank **134**.
(17) Further refer to FIG. **2**. The liquid cooling device **100** can utilize the first pump **110** to draw the hot water, exhausted from the cold plate **140**, from the third water tank **136** through corresponding water flow paths **139**, to reduce the temperature of the hot water with the heat radiating fins **138**, then entering into the first water tank **132**, passing through the first pump **110**, the first water tank **132** and corresponding water flow paths **139** to further reduce the temperature thereof, and then re-entering the cold plate **140** through the third water tank **136** so as to dissipate the heat of the electronic components such as graphics chips.
(18) The cold plate **140** includes a top cover **142** and a bottom plate **144**, and a first water inlet **201** and a first water outlet **202** are formed on the top cover **142**. The bottom plate **144** includes a first chamber **290** and first skived fins **260**.
(19) Similarly, the liquid cooling device **100** may further utilize the second pump **120** to draw the hot water, exhausted from the cold plate **140**, from the third water tank **136**, through the corresponding water flow paths **139**, to reduce the temperature of the hot water with the heat radiating fins **138**, then entering into the second water tank **134**, passing through the second pump **120**, the second water tank **134** and corresponding water flow paths **139** to further reduce the temperature thereof, and then re-entering the cold plate **140** through the third water tank **136** so as to dissipate the heat of the electronic components such as graphics chips.
(20) That is to say, the first pump **110** located on the left side may draw the working fluid from the third water tank **136** to the first water inlet **201** through the water flow paths **139** passing through in the heat radiating fins **138**, and draw the working fluid back from the first water outlet **202** to exchange the heat of the first skived fins **260** in the first chamber **290** so as to cool the electronic components such as graphics chips. In addition, the second pump **120** located on right side may draw the working fluid from the third water tank **136** to the first water inlet **201** through the water flow paths **139** passing through in the heat radiating fins **138**, and draw the working fluid back from the first water outlet **202** to simultaneously exchange the heat of the first skived fins **260** in the first chamber **290** so as to further cool the electronic components such as graphics chips. The first water tank **132**, the second water tank **134** and the third water tank **136** respectively include first partition plates **203** to respectively separate the first water tank **132**, the second water tank **134** and the third water tank **136** into two areas.
(21) Therefore, the liquid cooling device **100** may utilize the first pump **110** and the second pump **120** both disposed on the first surface **101** of the water cooling radiator **130** to cool the working fluid in two directions to effectively improve the heat dissipation efficiency of the liquid cooling device **100**. In addition, the first pump **110** and the second pump **120** may not influence the length and width of the water cooling radiator **130** because that the first pump **110** and the second pump **120** are both disposed on the first surface **101** of the water cooling radiator **130**.
(22) Referring to FIG. **3**, another embodiment of the liquid cooling device is illustrated. The liquid cooling device **300** includes a water cooling radiator **330**, a first pump **310** and a cold plate **340**. The cold plate **340** is disposed on a second surface **102** of the water cooling radiator **330**. In addition, the first pump **310** is disposed on a first surface **101** of the water cooling radiator **330**. The water cooling radiator **330** includes a first water tank **332**, a second water tank **334**, a third water tank **336**, a plurality of heat radiating fins **338** and a plurality of water flow paths **339**. The water flow paths **339** pass through in the heat radiating fins **338** and connect the first water tank **332**, the second water tank **334** and the third water tank **336**. The cold plate **340** and the first pump **310** are disposed on and communicate with the third water tank **336**, the working fluid is circulated through the first pump **310**, and the heat in the working fluid is transferred to the heat radiating fins **338**,

and then removed by the cooling fans.

(23) The cold plate **340** includes a top cover **342** and a bottom plate **344**, a first water inlet **301** and a first water outlet **302** formed on the top cover **342**. The bottom plate **344** includes a first chamber **390** and first skived fins **360**. In addition, the third water tank **336** includes a first partition plate **303** and a second partition plate **304** therein, and the first partition plate **303** and the second partition plate **304** separate the third water tank **336** into three areas.

(24) That is to say, the first pump **310** disposed on the third water tank **336** located in the middle may draw the working fluid from a corresponding area of the third water tank **336** through the water flow paths **339**, passing through in the heat radiating fins **338**, to the first water inlet **301** of the cold plate **340** by way of another corresponding area of the third water tank **336**, and then entering into the first chamber **390** to exchange the heat of the first skived fins **360** in the first chamber **390** so as to cool the electronic components such as graphics chips. Subsequently, the working fluid is transported to the first water outlet **302** of the cold plate **340**, and draw back into further another corresponding area, corresponding to the first water outlet **302**, of the third water tank **336**. In addition, the working fluid is directly transported to the first water tank **332** through the water flow paths **339**, passing through in the heat radiating fins **338**, and transported to the second water tank **334** through the water flow paths **339**, passing through in the heat radiating fins **338**, and then returns back to the third water tank **336** through the water flow paths **339**, passing through in the heat radiating fins **338**. Further, the working fluid pumped by the first pump **310** is then transported to the cold plate **340** again.

(25) That is to say, a first water inlet **301** and a first water outlet **302** of the cold plate **340** are respectively connected to two corresponding areas of the three areas of the third water tank **336**, and another area is only connected to the inlet of the first pump **310** and is not directly connected to the cold plate **340**. In addition, the first water inlet **301** of the cold plate **340** is communicated with the outlet of the first pump **310** through a corresponding area of the third water tank **336**.

(26) In addition, refer to FIG. **4**, further another embodiment of the liquid cooling device is illustrated. The liquid cooling device **400** includes a water cooling radiator **430**, a first pump **410**, a second pump **420**, a third pump **450** and a cold plate **440**. The cold plate **440** is disposed on a second surface **102** of the water cooling radiator **430**. The first pump **410**, the second pump **420** and the third pump **450** is disposed on the first surface **101** of the water cooling radiator **430**. In addition, the structure and function of the third pump **450** and the third water tank **436** are similar to those of the first pump **310** and the third water tank **336** as illustrated in FIG. **3**. The structure and function of the first water tank **432** and the first pump **410** are similar to those of the first water tank **132** and the first pump **110** as illustrated in FIG. **2**. The structure of the second water tank **434** and the second pump **420** is opposite to that of the first water tank **432** and the first pump **410**. In addition, the structure and function of the heat radiating fins **438**, the cold plate **440** and the water flow paths **439** are similar to those of the heat radiating fins **338**, the cold plate **340** and the water flow paths **339** as illustrated in FIG. **3**.

(27) That is to say, a first partition plate **403** and a second partition plate **404** are formed in the third water tank **436**, and the first partition plate **403** and the second partition plate **404** separate the third water tank **436** into three areas. In addition, the first water tank **432** and the second water tank **434** respectively include first partition plates **403** to respectively separate the first water tank **432** and the second water tank **434** into two areas.

(28) In addition, the cold plate **440** includes a top cover **442** and a bottom plate **444**. A first water inlet **401** and a first water outlet **402** are formed on the top cover **442**. The bottom plate **444** includes a first chamber **490** and first skived fins **460**.

(29) Therefore, the liquid cooling device **400** may simultaneously utilize three pumps to increase the flow velocity and flow rate of the working fluid, and the heat dissipation efficiency of the liquid cooling device **400**.

(30) Referring to FIG. **5** and FIG. **6**, still further another embodiment of the liquid cooling device is

illustrated. The liquid cooling device **500** includes a water cooling radiator **530**, a first pump **510**, a second pump **520** and a cold plate **540**. The cold plate **540** is disposed on the second surface **102** of the water cooling radiator **530**. The first pump **510** and the second pump **520** are disposed on the first surface **101** of the water cooling radiator **530**, but the present invention is not limited thereto.

(31) The first pump **510** is fixed and communicated with the first water tank **532**, and the second pump **520** is fixed and communicated with the second water tank **534**. The third water tank **536** includes a cross partition plate **550** to separate the third water tank **536** into four areas respectively connecting to corresponding first water inlet **501**, first water outlet **502**, second water inlet **503** and second water outlet **504** of the cold plate **540**. The cold plate **540** includes a top cover **542** and a bottom plate **544**. The first water inlet **501**, the first water outlet **502**, the second water inlet **503** and the second water outlet **504** are formed on the top cover **542**.

(32) In some embodiments, the cross partition plate **550** includes a first partition plate **551** and a second partition plate **552**. The second partition plate **552** is vertically cross-connected to the first partition plate **551** to form the cross partition plate **550**, and separate the third water tank **536** into four areas. In addition, the first water tank **532** and the second water tank **534** respectively include first partition plates **551** to respectively separate the first water tank **532** and the second water tank **534** into two areas.

(33) Referring to FIG. **6**, the bottom plate **544** includes a first chamber **690**, a second chamber **680**, a chamber partition plate **670**, first skived fins **660** and second skived fins **650**. The first chamber **690** includes a first water entrance channel **610** and a first water exit channel **620**, the second chamber **680** includes a second water entrance channel **630** and a second water exit channel **640** respectively corresponding to the first water inlet **501**, the first water outlet **502**, the second water inlet **503** and the second water outlet **504** of the top cover **542**.

(34) Therefore, the first water entrance channel **610**, the first water exit channel **620**, the second water entrance channel **630** and the second water exit channel **640** are respectively connected to corresponding areas of the third water tank **536**. The chamber partition plate **670** is formed between the first chamber **690** and the second chamber **680** to separate the first chamber **690** and the second chamber **680**, the first skived fins **660** are disposed in the first chamber **690** and located between the first water entrance channel **610** and the first water exit channel **620**, and the second skived fins **650** is disposed in the second chamber **680** and located between the second water entrance channel **630** and the second water exit channel **640**.

(35) Therefore, the first pump **510** located on the left side may transport the working fluid to the second water inlet **503** through the water flow paths **539**, passing through in the heat radiating fins **538**, and the working fluid is drawn back from the second water outlet **504** to exchange the heat of the second skived fins **650** in the second chamber **680** on the right side so as to cool the electronic components such as graphics chips. In addition, the second pump **520** located on the right side may transport the working fluid to the first water inlet **501**, and the working fluid is drawn back from the first water outlet **502** to exchange the heat of the first skived fins **660** of the first chamber **690** on the left side so as to cool the electronic components such as graphics chips. With two sets of skived fins, the heat dissipation efficiency of the liquid cooling device **500** can be further improved, and with two pumps located on the left and right sides, the heat exchange of the first skived fins **660** and the second skived fins **650** are performed respectively and may not interfere with each other and compete for the working fluid so as to provide more stable cooling capacity. In addition, the temperature distribution of the cold plate **540** may be adjusted to improve the cooling effect of the liquid cooling device **500** by adjusting the rotational speed and flow rate of the first pump **510** and the second pump **520** and/or the length and width of the first skived fins **660** and the second skived fins **650**.

(36) Referring to FIG. **7**, various configuration embodiments of cooling fans, water cooling radiators and pumps of the liquid cooling device are illustrated. The liquid cooling device may install the pumps on the first surface **101** or the second surface **102**, for example an air inlet surface

or air outlet surface of the water cooling radiator **130**, and the pumps are not installed on the side wall of the of the water cooling radiator **130** so as to provide a larger heat dissipation area of the water cooling radiator under a same length and width.

(37) It is worth noting that the liquid cooling device may be equipped with a plurality of cooling fans on the first surface **101** or the second surface **102** of the water cooling radiator **130** so that the heights of the cooling fans and the pumps are shared on the water cooling radiator **130** and the overall height of the liquid cooling device may not be increased.

(38) The configuration between the cooling fans, the water cooling radiator, the cold plate and the pumps of the liquid cooling device will be described in the following five configuration embodiments, but the present invention is not limited thereto.

(39) First, a first configuration embodiment **710** is illustrated, a liquid cooling device includes a first cooling fan **701** disposed adjacent to the first pump **110**, and a second cooling fan **702** disposed adjacent to the second pump **120**. The first pump **110** and the first cooling fan **701** are disposed on the first surface **101** of the water cooling radiator **130**, and the second pump **120** and the second cooling fan **702** are disposed on the first surface **101** of the water cooling radiator **130**. The cold plate **140** is disposed on the second surface **102** of the water cooling radiator **130**.

(40) Further, in a second configuration embodiment **720**, the first pump **110** and the first cooling fan **701** of the liquid cooling device are disposed on the second surface **102** of the water cooling radiator **130**, and the second pump **120** and the second cooling fan **702** are disposed on the second surface **102** of the water cooling radiator **130**. In addition, the cold plate **140** is disposed on the second surface **102** of the water cooling radiator **130**, and is communicated with an enlarged third water tank **136**. In addition, the volume of the third water tank **136** is greater than the volume of the first water tank **132** and the volume of the third water tank **136** is also greater than the volume of the second water tank **134** so that the storage volume of the liquid cooling device for storing the working fluid is increased without increasing the height of the liquid cooling device.

(41) In a third configuration embodiment **730** of the liquid cooling device, the first pump **110** and the first cooling fan **701** are disposed on the second surface **102** of the water cooling radiator **130**, and the second pump **120** and the second cooling fan **702** are disposed on the first surface **101** of the water cooling radiator **130**. In addition, the cold plate **140** is disposed on the second surface **102** of the water cooling radiator **130**, and is communicated with an enlarged third water tank **136**. In addition, the volume of the third water tank **136** is greater than the volume of the first water tank **132** and the volume of the third water tank **136** is also greater than the volume of the second water tank **134** so that the storage volume of the liquid cooling device for storing the working fluid is increased without increasing the height of the liquid cooling device. Furthermore, since the first pump **110** and the first cooling fan **701** are disposed on the second surface **102** of the water cooling radiator **130**, the second pump **120** and the second cooling fan **702** are disposed on the first surface **101** of the water cooling radiator **130**, and the left end and the right end of the water cooling radiator **130** includes a step therebetween so that the water cooling radiator **130** may match the components on various printed circuit board products with different appearances to improve the application field of the liquid cooling device.

(42) Furthermore, in a fourth configuration embodiment **740** of the liquid cooling device, the liquid cooling device includes a first cooling fan **701** disposed adjacent to the first pump **110**, a second cooling fan **702** disposed adjacent to the second pump **120**, and a third pump **760** fixed and communicated with the third water tank **436**. The first pump **110**, the first cooling fan **701**, the second pump **120**, the second cooling fan **702** and the third pump **760** are all disposed on the first surface **101** of the water cooling radiator **130**, and the first cooling fan **701** is disposed between the first pump **110** and the third pump **760** and the second cooling fan **702** is disposed between the second pump **120** and the third pump **760**. The cold plate **140** is disposed on the second surface **102** of the water cooling radiator **130**. Therefore, the liquid cooling device may increase the flow velocity, the flow rate, and the heat dissipation capacity of the working fluid with more pumps.

(43) Moreover, in a fifth configuration embodiment **750** of the liquid cooling device, the liquid cooling device is equipped with a single first pump **770** fixed and communicated with the third water tank **336**. In addition, the first pump **770** is disposed on the first surface **101** of the water cooling radiator **130** and located between the first cooling fan **701** and the second cooling fan **702**.
(44) With the above-mentioned various configuration embodiments of the liquid cooling device, the pump may not affect the length, width and height of the liquid cooling device so as to maximize the dimensions of the water cooling radiator and improve the heat dissipation efficiency of the liquid cooling device.
(45) In addition, FIG. **8** illustrates another embodiment of a water cooling radiator of a liquid cooling device. The water cooling radiator **810** includes a first water tank **812** and a second water tank **814**, a plurality of first heat radiating fins **816**, a plurality of second heat radiating fins **818** and a plurality of third heat radiating fins **820** are equipped between the first water tank **812** and the second water tank **814**. The water flow path **822**, the water flow path **824** and the water flow path **826** are respectively communicated with the first water tank **812** and the second water tank **814**, and disposed in the first heat radiating fins **816**, the second heat radiating fins **818** and the third heat radiating fins **820** to remove the heat of the working fluid flowing in the water flow path **822**, the water flow path **824** and the water flow path **826** with the first heat radiating fins **816**, the second heat radiating fins **818** and the third heat radiating fins **820**. It is worth noting that the second heat radiating fins **818** are formed between the first heat radiating fins **816** and the third heat radiating fins **820**, and the first heat radiating fins **816**, the second heat radiating fins **818** and the third heat radiating fins **820** respectively include different intervals.
(46) In some embodiments, the interval of the first heat radiating fins **816** is smaller than the interval of the second heat radiating fins **818**, and the interval of the second heat radiating fins **818** is smaller than the interval of the third heat radiating fins **820** so that the liquid cooling device may adjust the interval of the heat radiating fins to increase the heat dissipation efficiency of the liquid cooling device and reduce the working noise of the liquid cooling device according to the heat dissipation requirement of the liquid cooling device and the air intake of the cooling fan.
(47) Accordingly, the liquid cooling device of the present invention can provide a larger heat dissipation area of the water cooling radiator without increasing the length, width and height of the liquid cooling device, and further improve the heat dissipation efficiency of the liquid cooling device.
(48) As is understood by a person skilled in the art, the foregoing preferred embodiments of the present invention are illustrative of the present invention rather than limiting of the present invention. It is intended that various modifications and similar arrangements be included within the spirit and scope of the appended claims, the scope of which should be accorded the broadest interpretation so as to encompass all such modifications and similar structures.

Claims

1. A liquid cooling device, comprising: a water cooling radiator, comprising a first surface and a second surface, the first surface and the second surface respectively located on opposite sides of the water cooling radiator; a first pump disposed on the first surface or the second surface of the water cooling radiator; a cold plate disposed on the second surface of the water cooling radiator, wherein the water cooling radiator further comprises: a first water tank; a second water tank; a third water tank; a plurality of heat radiating fins disposed between the first water tank, the second water tank and the third water tank; a plurality of water flow paths passing through in the heat radiating fins; and at least one cooling fan disposed on the first surface or the second surface of the water cooling radiator, and located adjacent to the first pump.
2. The liquid cooling device of claim 1, further comprising a second pump, wherein the first pump is fixed and communicated with the first water tank and the second pump is fixed and

communicated with the second water tank.
3. The liquid cooling device of claim 2, further comprising: a first cooling fan disposed on the first surface or the second surface of the water cooling radiator, and located adjacent to the first pump; and a second cooling fan disposed on the first surface or the second surface of the water cooling radiator, and located adjacent to the second pump.
4. The liquid cooling device of claim 3, wherein the first pump and the first cooling fan are disposed on the first surface of the water cooling radiator, and the second pump and the second cooling fan are also disposed on the first surface of the water cooling radiator.
5. The liquid cooling device of claim 3, wherein the first pump and the first cooling fan are disposed on the second surface of the water cooling radiator, and the second pump and the second cooling fan are also disposed on the second surface of the water cooling radiator.
6. The liquid cooling device of claim 3, wherein the first pump and the first cooling fan are disposed on the second surface of the water cooling radiator, and the second pump and the second cooling fan are disposed on the first surface of the water cooling radiator.
7. The liquid cooling device of claim 6, wherein two ends of the water cooling radiator comprises a step.
8. The liquid cooling device of claim 3, further comprising a third pump fixed and communicated with the third water tank.
9. The liquid cooling device of claim 8, wherein the first pump, the first cooling fan, the second pump, the second cooling fan and the third pump are all disposed on the first surface of the water cooling radiator, and the first cooling fan is located between the first pump and the third pump, and the second cooling fan is located between the second pump and the third pump.
10. The liquid cooling device of claim 1, wherein the first pump is fixed and communicated with the third water tank.
11. The liquid cooling device of claim 10, wherein the third water tank comprises: a first partition plate; and a second partition plate, wherein the second partition plate and the first partition plate separate the third water tank into three areas, wherein a first water inlet and a first water outlet of the cold plate are respectively connected to two corresponding areas of the three areas, and another area of the three areas is connected to the first pump and disconnected from the cold plate.
12. The liquid cooling device of claim 10, wherein the first pump is disposed on the first surface of the water cooling radiator, and located between a first cooling fan and a second cooling fan.
13. The liquid cooling device of claim 1, wherein the heat radiating fins comprise: a plurality of first heat radiating fins; a plurality of second heat radiating fins; and a plurality of third heat radiating fins, wherein the second heat radiating fins are formed between the first heat radiating fins and the third heat radiating fins, and the first heat radiating fins, the second heat radiating fins and the third heat radiating fins have different intervals.
14. The liquid cooling device of claim 13, wherein an interval of the first heat radiating fins is smaller than an interval of the second heat radiating fins, and the interval of the second heat radiating fins is smaller than an interval of the third heat radiating fins.
15. The liquid cooling device of claim 1, wherein the cold plate is connected to the third water tank.
16. The liquid cooling device of claim 15, wherein the third water tank comprises a first partition plate to separate the third water tank into two areas.
17. The liquid cooling device of claim 16, wherein the cold plate comprises: a first chamber comprising a first water inlet and a first water outlet, wherein the first water inlet and the first water outlet are respectively connected to corresponding areas of the two areas of the third water tank; and first skived fins disposed in the first chamber, and located between the first water inlet and the first water outlet.
18. The liquid cooling device of claim 17, wherein the third water tank further comprises a second partition plate, and the second partition plate and the first partition plate are formed a cross

partition plate to separate the third water tank into four areas.

19. The liquid cooling device of claim 18, wherein the cold plate further comprises: a second chamber comprising a second water inlet and a second water outlet, wherein the first water inlet, the first water outlet, the second water inlet and the second water outlet are respectively connected to corresponding areas of the four areas of the third water tank; a chamber partition plate formed between the first chamber and the second chamber to isolate the first chamber and the second chamber; and second skived fins disposed in the second chamber, and located between the second water inlet and the second water outlet.